

NE523 Homework 1

Kyle Hansen

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Question 1 Two-species Boltzmann Equation

The Boltzmann equation can be derived by computing the transformation that a small element in phase space sees in a short time δt . From section 2, in the absence of collisions and considering only first-order δt terms, the particles at $(\mathbf{r}, \mathbf{v})$ will stream to $\left(\mathbf{r} + \mathbf{v}\delta t, \mathbf{v} + \frac{\mathbf{F}|_{t=t_0}}{m}\delta t, t_0 + \delta t\right)$ after a short time δt , or

$$f\left(\mathbf{r} + \mathbf{v}\delta t, \mathbf{v} + \frac{\mathbf{F}|_{t=t_0}}{m}\delta t, t_0 + \delta t\right) = f(\mathbf{r}, \mathbf{v}, t_0). \quad (1)$$

Note that δt must be small enough to neglect the higher-order terms. The effect of scattering can be denoted with $\left(\frac{\partial f}{\partial t}\right)_{coll}$, the net change in f over time due to scattering (into and out of $(\mathbf{r}, \mathbf{v})$). Then in time δt the total change in f is $\left(\frac{\partial f}{\partial t}\right)_{coll} \delta t$, and

$$f\left(\mathbf{r} + \mathbf{v}\delta t, \mathbf{v} + \frac{\mathbf{F}|_{t=t_0}}{m}\delta t, t_0 + \delta t\right) = f(\mathbf{r}, \mathbf{v}, t_0) + \left(\frac{\partial f}{\partial t}\right)_{coll} \delta t. \quad (2)$$

Expanding the left-hand side as a Taylor series in t , this becomes

$$\cancel{f(\mathbf{r}, \mathbf{v}, t)} + (\nabla_r f \cdot \mathbf{v}) \delta t + \left(\nabla_v f \cdot \frac{\mathbf{F}}{m}\right) \delta t + \left(\frac{\partial f}{\partial t}\right) \delta t = \cancel{f(\mathbf{r}, \mathbf{v}, t)} + \left(\frac{\partial f}{\partial t}\right)_{coll} \delta t \quad (3)$$

$$\left(\mathbf{v}\nabla_r + \frac{\mathbf{F}}{m}\nabla_v + \frac{\partial}{\partial t}\right) f(\mathbf{r}, \mathbf{v}, t) \delta t = \left(\frac{\partial f}{\partial t}\right)_{coll} \delta t \quad (4)$$

$$\left(\mathbf{v}\nabla_r + \frac{\mathbf{F}}{m}\nabla_v + \frac{\partial}{\partial t}\right) f(\mathbf{r}, \mathbf{v}, t) = \left(\frac{\partial f}{\partial t}\right)_{coll}. \quad (5)$$

The same applies for particles of species g , so

$$\left(\mathbf{v} \nabla_r + \frac{\mathbf{F}}{m} \nabla_v + \frac{\partial}{\partial t} \right) f(\mathbf{r}, \mathbf{v}, t) = \left(\frac{\partial f}{\partial t} \right)_{coll} \quad (6)$$

$$\left(\mathbf{v} \nabla_r + \frac{\mathbf{G}}{m} \nabla_v + \frac{\partial}{\partial t} \right) g(\mathbf{r}, \mathbf{v}, t) = \left(\frac{\partial g}{\partial t} \right)_{coll}. \quad (7)$$

The collision integral can be found by computing the rate of scattering into or out of $(\mathbf{r}, \mathbf{v})$ due to collisions of particles of both species,

$$\left(\frac{\partial f}{\partial t} \right)_{coll} = \sum_{i=f,g} R_{gain,f,i} - R_{loss,f,i}. \quad (8)$$

The rate of collisions with a beam of species i and velocity $\mathbf{v}_i$, which cause an f -particle of velocity $\mathbf{v}$ to scatter to velocity $\mathbf{v}'$, is given by

$$I_i \sigma_{f,i}(\mathbf{r}, \mathbf{v}_i, \mathbf{v} \rightarrow \mathbf{v}', \mathbf{v}'), \quad (9)$$

or, equivalently, with a change in notation,

$$I_i \sigma_{f,i}(\mathbf{r}, \mathbf{v}_i', \mathbf{v}' \rightarrow \mathbf{v}_i, \mathbf{v}), \quad (10)$$

where $\sigma_{f,i}$ is the equivalent cross-sectional area (accounting for quantum effects) of a collision between an f -particle and i -particle (where $i = f, g$). Then the total rate (rate-density times density of f -particles) of these collisions is

$$f(\mathbf{r}, \mathbf{v}, t) I_i \sigma_{f,i}(\mathbf{r}, \mathbf{v}_i, \mathbf{v} \rightarrow \mathbf{v}', \mathbf{v}'), \quad (11)$$

or

$$f(\mathbf{r}, \mathbf{v}', t) I_i \sigma_{f,i}(\mathbf{r}, \mathbf{v}_i', \mathbf{v}' \rightarrow \mathbf{v}_i, \mathbf{v}). \quad (12)$$

The intensity of a particle beam is density times its speed (relative to the target, in this case the f -particle), or

$$I_i(\mathbf{r}, \mathbf{v}_i, t) = i(\mathbf{r}, \mathbf{v}_i, t) |\mathbf{v}_i - \mathbf{v}| \quad (13)$$

then the total rate of collisions with i -particles of velocity $\mathbf{v}_i$ which cause an f -particle of velocity $\mathbf{v}$ to scatter to $\mathbf{v}'$ is

$$f(\mathbf{r}, \mathbf{v}, t) i(\mathbf{r}, \mathbf{v}_i, t) |\mathbf{v}_i - \mathbf{v}| \sigma_{f,i}(\mathbf{r}, \mathbf{v}_i, \mathbf{v} \rightarrow \mathbf{v}', \mathbf{v}'). \quad (14)$$

The total removal rate is this function, integrated over all $\mathbf{v}', \mathbf{v}_i, \mathbf{v}_i'$, since a collision with any result, and with an i -particle of any energy will cause particle removal. So,

$$R_{loss,f,f} = \int d^3v' \int d^3v_f \int d^3v'_f f(\mathbf{r}, \mathbf{v}, t) f(\mathbf{r}, \mathbf{v}_f, t) |\mathbf{v}_f - \mathbf{v}| \sigma_{f,f}(\mathbf{r}, \mathbf{v}_f, \mathbf{v} \rightarrow \mathbf{v}_f', \mathbf{v}') \quad (15)$$

$$R_{loss,f,g} = \int d^3v' \int d^3v_g \int d^3v'_g f(\mathbf{r}, \mathbf{v}, t) g(\mathbf{r}, \mathbf{v}_g, t) |\mathbf{v}_g - \mathbf{v}| \sigma_{f,g}(\mathbf{r}, \mathbf{v}_g, \mathbf{v} \rightarrow \mathbf{v}_g', \mathbf{v}'), \quad (16)$$

with similar rates for g . The gain rate is similarly found by integrating over all initial particle velocities $\mathbf{v}'$, $\mathbf{v}'_i$ and all resulting $\mathbf{v}_i$, since a collision at any velocity, with a second particle of any velocity may lead to scattering to $\mathbf{v}$. So,

$$R_{gain,f,f} = \int d^3v' \int d^3v'_f \int d^3v_f f(\mathbf{r}, \mathbf{v}, t) f(\mathbf{r}, \mathbf{v}_f, t) |\mathbf{v}_f - \mathbf{v}| \sigma_{f,f}(\mathbf{r}, \mathbf{v}_f', \mathbf{v}' \rightarrow \mathbf{v}_f, \mathbf{v}) \quad (17)$$

$$R_{gain,f,g} = \int d^3v' \int d^3v'_f \int d^3v_g f(\mathbf{r}, \mathbf{v}, t) g(\mathbf{r}, \mathbf{v}_g, t) |\mathbf{v}_g - \mathbf{v}| \sigma_{f,g}(\mathbf{r}, \mathbf{v}_g', \mathbf{v}' \rightarrow \mathbf{v}_g, \mathbf{v}) \quad (18)$$

with similar rates for g . Then the system that describes the population is

$$\begin{aligned} & \left(\mathbf{v} \nabla_r + \frac{\mathbf{F}}{m} \nabla_v + \frac{\partial}{\partial t} \right) f(\mathbf{r}, \mathbf{v}, t) \\ &= \sum_{i=f,g} \int d^3v' \int d^3v'_i \int d^3v_i v_i f(\mathbf{r}, \mathbf{v}, t) i(\mathbf{r}, \mathbf{v}_i, t) |\mathbf{v}_i - \mathbf{v}| \sigma_{f,i}(\mathbf{r}, \mathbf{v}_i', \mathbf{v}' \rightarrow \mathbf{v}_f, \mathbf{v}) \\ & \quad - \sum_{i=f,g} \int d^3v' \int d^3v_i \int d^3v'_f f(\mathbf{r}, \mathbf{v}, t) i(\mathbf{r}, \mathbf{v}_i, t) |\mathbf{v}_i - \mathbf{v}| \sigma_{f,i}(\mathbf{r}, \mathbf{v}_i, \mathbf{v} \rightarrow \mathbf{v}_i', \mathbf{v}') \end{aligned} \quad (19)$$

$$\begin{aligned} & \left(\mathbf{v} \nabla_r + \frac{\mathbf{G}}{m} \nabla_v + \frac{\partial}{\partial t} \right) g(\mathbf{r}, \mathbf{v}, t) \\ &= \sum_{i=f,g} \int d^3v' \int d^3v'_i \int d^3v_i v_i g(\mathbf{r}, \mathbf{v}, t) i(\mathbf{r}, \mathbf{v}_i, t) |\mathbf{v}_i - \mathbf{v}| \sigma_{g,i}(\mathbf{r}, \mathbf{v}_i', \mathbf{v}' \rightarrow \mathbf{v}_g, \mathbf{v}) \\ & \quad - \sum_{i=f,g} \int d^3v' \int d^3v_i \int d^3v'_g g(\mathbf{r}, \mathbf{v}, t) i(\mathbf{r}, \mathbf{v}_i, t) |\mathbf{v}_i - \mathbf{v}| \sigma_{g,i}(\mathbf{r}, \mathbf{v}_i, \mathbf{v} \rightarrow \mathbf{v}_i', \mathbf{v}') \end{aligned} \quad (20)$$

Question 2 Differential Volume Invariant

By Cavalieri's principle, two objects with the same height and constant cross-sectional area have the same volume (see Figure 1 for illustration). Generalized to higher dimensions, if two n -dimensional volumes have the same height (for example, length when projected onto the v_x axis) and cross-sectional volume across the other $(n - 1)$ dimensions, then those two volumes are equal.

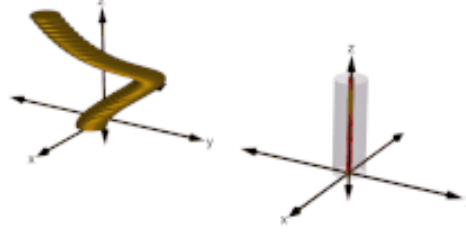


Figure 1: Cavalieri's principle [Wikipedia]

Consider the 6-dimensional prism confined by opposing corners $(x_0, y_0, z_0, v_{0,x}, v_{0,y}, v_{0,z})$ and $((x_0 + \Delta x), \dots, (v_{0,x} + \Delta v_x), \dots)$ and its change in time Δt . The Taylor series about t_0 for a particle's position and velocity are given by

$$x(t) = x_0 + \frac{\partial x}{\partial t}(\Delta t) + \frac{\partial^2 x}{\partial t^2}(\Delta t)^2 + \mathcal{O}(\Delta t^3) \quad (21a)$$

$$y(t) = \dots \quad (21b)$$

$$z(t) = \dots \quad (21c)$$

$$v_x(t) = v_{0,x} + \frac{\partial v_x}{\partial t}(\Delta t) + \frac{\partial^2 v_x}{\partial t^2}(\Delta t)^2 + \mathcal{O}(\Delta t^3) \quad (21d)$$

$$v_y(t) = \dots \quad (21e)$$

$$v_z(t) = \dots \quad (21f)$$

Ignoring second order terms in Δt , and using $\frac{\partial x}{\partial t} = v_x$ and $\frac{\partial v_x}{\partial t} = \frac{F_x}{m}$, this becomes

$$x(t) = x_0 + v_x(\Delta t) \quad (22a)$$

$$y(t) = \dots \quad (22b)$$

$$z(t) = \dots \quad (22c)$$

$$v_x(t) = v_{0,x} + \frac{F_x}{m}(\Delta t) \quad (22d)$$

$$v_y(t) = \dots \quad (22e)$$

$$v_z(t) = \dots \quad (22f)$$

Then, consider the projection of the volumes (at t_0 and $t_0 + \Delta t$) onto the x - v_x plane, with corners $\{(x_1, v_{x,1}), (x_2, v_{x,2}), (x_3, v_{x,3}), (x_4, v_{x,4})\}$ (the “differential” area at t_0) and the area $\{(x'_1, v'_{x,1}), (x'_2, v'_{x,2}), (x'_3, v'_{x,3}), (x'_4, v'_{x,4})\}$ (the “differential” area at $t_0 + \Delta t$), as shown in Figure 2.

Corners 1 and 2 see the same x -transformation of $v_0 \Delta t$, and corners 3 and 4 undergo the same x -transformation of $(v_{0,x} + \Delta v_x) \Delta t$. Likewise, corners 1 and 4 undergo the same

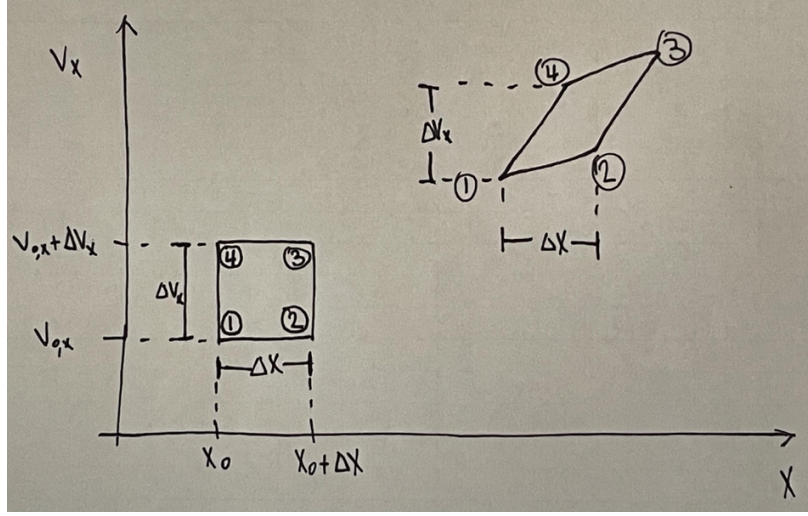


Figure 2: Projected differential volume transformation

v_x -transformation of $\frac{F_x(x_0)}{m}\Delta t$ and corners 2 and 3 undergo the same v_x -transformation of $\frac{F_x(x_0+\Delta x)}{m}\Delta t$. Then the transformed shape is a parallelogram with base Δx and height Δv_x , the same as the original shape; the two shapes have the same area. The same applies for the projections onto y - v_y and z - v_z .

Since x and v_x do not depend on the y and z terms, the area of the x - v_x plane is constant for any (y, z, v_y, v_z) , so by Cavalieri's principle the volume can be determined by calculating $A_{x,v_x} \cdot V_{y,z,v_y,v_z}$. Applying this again to the (y, v_y) plane, the volume can be determined by calculating $A_{x,v_x} A_{y,v_y} A_{z,v_z}$ —since each of these areas remain constant after Δt , the total 6-dimensional volume of the shape remains the same.

Question 3 Maxwell-Boltzmann Moments

The distribution with arbitrary bulk velocity $\mathbf{v}_0$ is

$$f_0(\mathbf{v}) = n \left(\frac{m}{2\pi kT} \right)^{3/2} \exp \left[-\frac{m|\mathbf{v} - \mathbf{v}_0|^2}{2kT} \right]. \quad (23)$$

Let $\mathbf{w} \stackrel{\text{def}}{=} \mathbf{v} - \mathbf{v}_0$. Then distribution in the transformed coordinate is

$$f_0(\mathbf{w} + \mathbf{v}_0) = n \left(\frac{m}{2\pi kT} \right)^{3/2} \exp \left[-\frac{m|\mathbf{w}|^2}{2kT} \right]. \quad (24)$$

Then the transformation disappears when integrating the zeroth moment, since the bounds of integration are infinite: $\int d^3w f_0(\mathbf{w} + \mathbf{v}_0) = \int d^3w f_0(\mathbf{w})$. Then the zeroth moment is

$$\int d^3w f_0(\mathbf{w}) = \int d^3w f_0(\mathbf{w} + \mathbf{v}_0) = \int d^3w n \left(\frac{m}{2\pi kT} \right)^{3/2} \exp \left[-\frac{m|\mathbf{w}|^2}{2kT} \right]. \quad (25)$$

Transforming to spherical coordinates, the differential volume element becomes $d^3w = w^2 \sin \theta d\theta d\phi dw$, where $0 \leq \theta \leq \pi$ is the polar angle, $0 \leq \phi \leq 2\pi$ is the azimuthal angle, and $0 \leq (w = |\mathbf{w}|) < \infty$ is the speed, and the zeroth moment is

$$\int d^3w f_0(\mathbf{w}) = \int_0^\pi \sin \theta d\theta \int_0^{2\pi} d\phi \int_0^\infty n \left(\frac{m}{2\pi kT} \right)^{3/2} \exp \left[-\frac{mw^2}{2kT} \right] w^2 dw \quad (26a)$$

$$= 4\pi n \left(\frac{m}{2\pi kT} \right)^{3/2} \int_0^\infty \exp \left[-\frac{mw^2}{2kT} \right] w^2 dw. \quad (26b)$$

With a change of variables $t = \frac{mw^2}{2kT}$, this becomes

$$\int d^3w f_0(\mathbf{w}) = 4\pi n \left(\frac{m}{2\pi kT} \right)^{3/2} \int_0^\infty \frac{1}{2} \left(\frac{2kT}{m} \right)^{3/2} t^{1/2} \exp[-t] dt. \quad (27)$$

This integral can be evaluated using the gamma function $\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt$,

$$\int d^3w f_0(\mathbf{w}) = 4\pi n \left(\frac{m}{2\pi kT} \right)^{3/2} \frac{1}{2} \left(\frac{2kT}{m} \right)^{3/2} \Gamma \left(\frac{3}{2} \right) \quad (28)$$

$$= 4\pi n \left(\frac{m}{2\pi kT} \right)^{3/2} \frac{1}{2} \left(\frac{2kT}{m} \right)^{3/2} \frac{\sqrt{\pi}}{2}, \quad (29)$$

so the zeroth moment is

$$\boxed{\int d^3w f_0(\mathbf{w}) = n}. \quad (30)$$

The first moment is

$$\begin{aligned} \int d^3w (\mathbf{w} + \mathbf{v}_0) f_0(\mathbf{w} + \mathbf{v}_0) &= \int d^3w (\mathbf{w} + \mathbf{v}_0) f_0(\mathbf{w} + \mathbf{v}_0) \\ &= \int d^3w \mathbf{w} f_0(\mathbf{w} + \mathbf{v}_0) + \int d^3w \mathbf{v}_0 f_0(\mathbf{w} + \mathbf{v}_0). \end{aligned} \quad (31)$$

The second part of this integral can be evaluated easily using the zeroth moment, since $\mathbf{v}_0$ does not depend on $\mathbf{w}$; $\int d^3w \mathbf{v}_0 f_0(\mathbf{w} + \mathbf{v}_0) = \mathbf{v}_0 n$. The first part of this integral evaluates to zero, since the bracketed term in

$$\begin{aligned} \int d^3w \mathbf{w} f_0(\mathbf{w} + \mathbf{v}_0) \\ = n \left(\frac{m}{2\pi kT} \right)^{3/2} \int_0^\infty \exp \left(-\frac{mw^2}{2kT} \right) w^3 dw \left[\int_0^\pi \sin \theta d\theta \int_0^{2\pi} d\phi \hat{w}(\theta, \phi) \right], \end{aligned} \quad (32)$$

where $\hat{w}(\theta, \phi)$ is the unit vector that points in the direction (θ, ϕ) , is equal to zero due to spherical symmetry.

Then the first moment is

$$\boxed{\int d^3v \mathbf{v} f_0(\mathbf{v}) = n \mathbf{v}_0}. \quad (33)$$

The second moment is

$$\int d^3v \left(\frac{1}{2} m v^2 \right) f_0(\mathbf{v}). \quad (34)$$

Applying the same transformation $\mathbf{w} = \mathbf{v} - \mathbf{v}_0$, this can be written

$$\int d^3v \left(\frac{1}{2} m v^2 \right) f_0(\mathbf{v}) = n \left(\frac{m}{2\pi kT} \right)^{3/2} \int d^3w \left(\frac{1}{2} m |\mathbf{w} + \mathbf{v}_0|^2 \right) \exp \left[\frac{-mw^2}{2kT} \right], \quad (35)$$

where

$$|\mathbf{w} + \mathbf{v}_0|^2 = (w_x + v_{0,x})^2 + (w_y + v_{0,y})^2 + (w_z + v_{0,z})^2 \quad (36a)$$

$$= (w_x^2 + 2w_x v_{0,x} + v_{0,x}^2) + (w_y^2 + 2w_y v_{0,y} + v_{0,y}^2) + (w_z^2 + 2w_z v_{0,z} + v_{0,z}^2) \quad (36b)$$

Then, translating to spherical coordinates, this becomes

$$\begin{aligned} |\mathbf{w} + \mathbf{v}_0|^2 = & (w_x^2 + 2 \sin \theta \cos \phi \cdot w v_{0,x} + v_{0,x}^2) \\ & + (w_y^2 + 2 \sin \theta \sin \phi \cdot w v_{0,y} + v_{0,y}^2) \\ & + (w_z^2 + 2 \cos \phi \cdot w v_{0,z} + v_{0,z}^2). \end{aligned} \quad (37)$$

Then, rewriting the integral in spherical coordinates yields

$$\begin{aligned} & n \left(\frac{m}{2\pi kT} \right)^{3/2} \left[\int_0^\pi \sin \theta d\theta \int_0^{2\pi} d\phi \int_0^\infty dw \left(\frac{1}{2} m w^4 \right) \exp \left(\frac{-mw^2}{2kT} \right) \right. \\ & + \int_0^\pi d\theta \int_0^{2\pi} d\phi \int_0^\infty dw \left(m (\sin^2 \theta \cos \phi v_{0,x} + \sin^2 \theta \sin \phi v_{0,y} + \sin \theta \cos \phi v_{0,z}) w^3 \right) \exp \left(\frac{-mw^2}{2kT} \right) \\ & \left. + \frac{1}{2} m v_0^2 \int_0^\pi \sin \theta d\theta \int_0^{2\pi} d\phi \int_0^\infty dw (w^2) \exp \left(\frac{-mw^2}{2kT} \right) \right]. \end{aligned} \quad (38)$$

The second line of the bracketed term evaluates to zero since $\int_0^{2\pi} \sin \phi d\phi = \int_0^{2\pi} \cos \phi d\phi = 0$, so this equation becomes

$$\frac{1}{2}m \cdot n \left(\frac{m}{2\pi kT} \right)^{3/2} \cdot 4\pi \int_0^\infty dw (v_0^2 w^2 + w^4) \exp \left(-\frac{mw^2}{2kT} \right). \quad (39)$$

With the change of variables $t = \frac{mw^2}{2kT}$, this becomes

$$4\pi \frac{1}{2}m \cdot n \left(\frac{m}{2\pi kT} \right)^{3/2} \int_0^\infty dt \frac{1}{2} \left(\frac{2kT}{m} \right)^{1/2} t^{-1/2} \left[\left(\frac{2kT}{m} \right)^2 t^2 + v_0^2 \frac{2kT}{m} t \right] \exp(-t) \quad (40a)$$

$$4\pi(\pi)^{-3/2} \frac{1}{4}m \cdot n \left(\frac{m}{2kT} \right) \int_0^\infty dt \left[\left(\frac{2kT}{m} \right)^2 t^{3/2} + v_0^2 \frac{2kT}{m} t^{1/2} \right] \exp(-t) \quad (40b)$$

$$\frac{1}{\sqrt{\pi}}m \cdot n \int_0^\infty dt \left[\left(\frac{2kT}{m} \right) t^{3/2} + v_0^2 t^{1/2} \right] \exp(-t) \quad (40c)$$

$$\frac{mn}{\sqrt{\pi}} \left[\frac{2kT}{m} \Gamma \left(\frac{5}{2} \right) + v_0^2 \Gamma \left(\frac{3}{2} \right) \right] \quad (40d)$$

$$\frac{mn}{\sqrt{\pi}} \left[\frac{2kT}{m} \left(\frac{3\sqrt{\pi}}{4} \right) + v_0^2 \left(\frac{\sqrt{\pi}}{2} \right) \right], \quad (40e)$$

so the second moment is

$$\boxed{\int d^3v \left(\frac{1}{2}mv^2 \right) f_0(\mathbf{v}) = n \left(\frac{3}{2}kT + \frac{1}{2}mv_0^2 \right)}. \quad (41)$$